

A Combined Time-Frequency and Common Spatial-Spectral Pattern Approach for EEG-Based Motor Imagery Classification

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Abstract— Brain-Computer Interfaces (BCIs) are revolutionizing neurorehabilitation, providing crucial communication and control for individuals with severe motor impairments from conditions like ALS, spinal cord injuries, or stroke. By creating direct links between brain activity and external devices, BCIs bypass damaged neural pathways, thus restoring motor function and significantly enhancing quality of life. Electroencephalography (EEG) is a favored BCI modality due to its accessibility and cost-effectiveness. However, a major challenge lies in the substantial impact of cognitive and individual differences on motor imagery (MI) task performance and overall BCI accuracy. This research introduces a novel method to overcome these challenges, focusing on enhanced MI classification. Our approach synergistically integrates Common Spatial-Spectral Pattern filters with the Tunable-Q Wavelet Transform. This powerful combination was applied to the extensive CHO-2017 database (52 participants), which uniquely captures significant inter-individual cognitive variations, specifically to distinguish between left and right-hand MI tasks. A critical aspect of our method is the utilization of only the top 10 most discriminative features extracted through this hybrid technique. This deliberate streamlining maximizes classification efficacy while maintaining computational efficiency. This tailored feature set demonstrated remarkable effectiveness, performing across 99% of participants. When integrated with a K-Nearest Neighbors classifier, this approach achieved an outstanding accuracy of 98.84%, notably surpassing existing state-of-the-art methods in the field. These findings hold significant promise for developing more accurate and robust BCI systems capable of extracting optimal commands for diverse MI applications, ultimately advancing neurorehabilitation outcomes.

Keywords—Motor Imagery Tasks, Common Spatial-Spectral Patterns, Tunable-Q Wavelet Transform

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) represent a pivotal technological advancement that offers individuals with severe motor impairments the prospect of restoring lost communication abilities, regaining control over their environment, and enhancing their overall quality of life. These systems establish a direct communication pathway between the brain and external devices, bypassing traditional motor pathways. Among the various BCI modalities, electroencephalography (EEG) has emerged as a particularly attractive option due to its non-invasive nature, affordability, portability, and relatively high temporal resolution. EEG's ease of use and accessibility make it a practical solution for real-world applications in neurorehabilitation and assistive

technology. At the heart of many EEG-based BCIs lies the principle of decoding neural activity associated with Motor Imagery (MI)—the mental simulation of movement without actual physical execution. During MI, specific brain regions associated with motor planning and execution become activated, resulting in distinct changes in electrophysiological signals, primarily observed as modulations in the mu (8-12 Hz) and beta (13-30 Hz) frequency bands over the sensorimotor cortex. These changes can be detected and analyzed to infer the user's intended movement.

Despite the considerable progress made in the field of MI-BCIs, achieving consistently high levels of accuracy, robustness, and usability remains a significant challenge. Several factors contribute to this difficulty, including the inherent variability in brain activity patterns across individuals, the influence of cognitive factors such as attention and motivation, and the presence of noise and artifacts in EEG recordings. Inter-individual variability, in particular, poses a major hurdle, as the neural correlates of MI can differ substantially from one person to another, making it difficult to develop generic BCI models that generalize well across users. To address these challenges, researchers have explored a wide range of signal processing and machine learning techniques aimed at extracting relevant features from EEG data and building robust classification models.

Classical machine learning approaches have historically formed the foundation of MI-BCI research. Studies from 2021-2022 have focused on enhancing feature extraction and selection methodologies. For example, Mohedival et al. [1, 2] investigated the use of Discrete Wavelet Transform (DWT) for decomposing EEG signals into different frequency sub-bands, followed by the extraction of statistical and band power features. They then employed sophisticated feature selection algorithms, such as Harmony Search, mutual information, chi-square, and correlation, to identify the most relevant features for MI classification. These methods were evaluated on the BNCI Horizon 2020 database, a widely used benchmark dataset in the BCI community. Blanco Diaz et al. [3] explored various implementations of Common Spatial Patterns (CSP), including Filter Bank CSP (FBCSP) and Filter Bank Sub-Band CSP (FBCSSP), focusing on spectro-spatial analysis to identify discriminative EEG patterns. Their research utilized both the Gigascience database and the BCI Competition IVa dataset. Seddiq et al. [4] proposed the use of Modified Variational Mode Decomposition (MVMD) for feature extraction, combined with Maximum Noise Fraction (MNF) Principal Component Analysis (PCA) for noise reduction.

Taran et al. [5] investigated the application of the Tunable Q Wavelet Transform (TQWT) for MI classification,

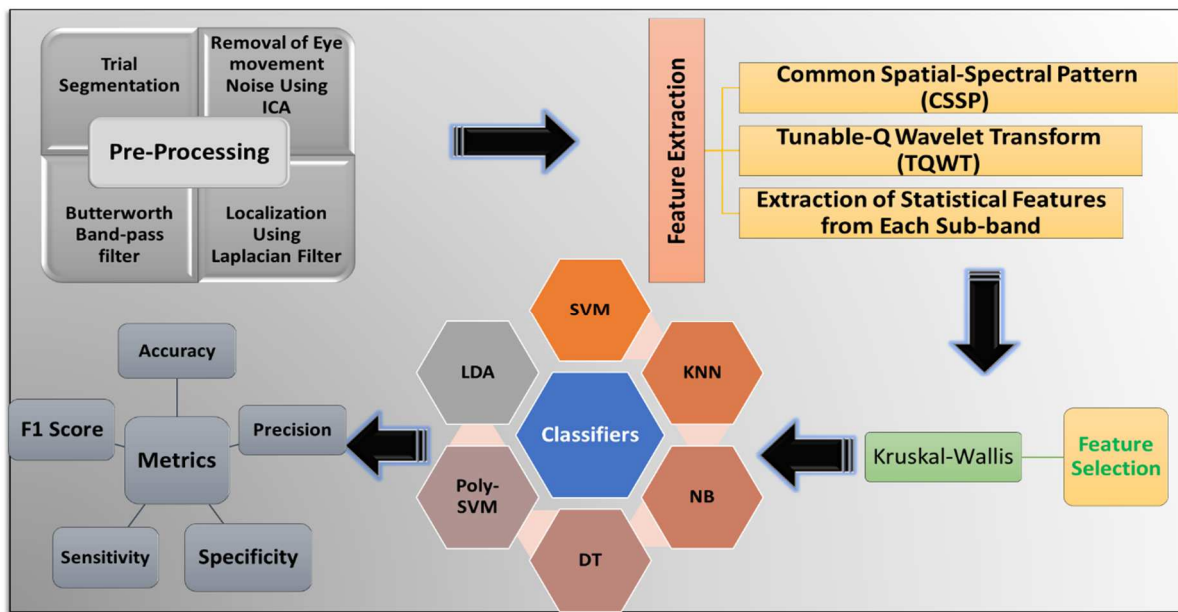


Fig. 1. Block Diagram of the Proposed Method

demonstrating its effectiveness on the BCI-Competition III and IVa datasets.

The field has experienced a surge in the application of deep learning (DL) approaches, which have shown remarkable potential for capturing complex spatio-temporal dynamics within EEG signals. Early studies (2017-2019) explored the use of Convolutional Neural Networks (CNNs) for MI classification. Tang et al. [6] developed individual CNNs tailored for hand MI classification. Zhou et al. [7] used spectral images derived from Short-Time Fourier Transforms (STFT) as input for a VGG-16 CNN, while Alazari et al. [8] employed time-frequency images generated using the Cho-William distribution as input for CNNs. From 2021 to 2024, advancements have included deep learning networks with transfer learning for multi-database EEG analysis, including data from stroke patients [9]. Innovations in deep geometric neural networks [10], supervised contrast learning for domain generalization [11], and neighborhood-preserving transfer learning in Riemannian space [12] have pushed the boundaries of robust EEG decoding. Hybrid DL models, such as the combination of 3D CNNs and Generative Adversarial Networks (GANs) utilizing 3D time-frequency-fuzzy feature maps, have been employed to improve MI classification accuracy by generating synthetic data [13].

Despite these advancements, the persistent challenge of inter-individual variability continues to limit the widespread adoption of MI-BCI technology. Addressing this issue is crucial for developing BCI systems that can be effectively used by a diverse population of users. This paper introduces a novel hybrid methodology designed to specifically mitigate the effects of inter-individual variability. Our approach as demonstrated in Fig. 1, integrates Common Spatial-Spectral Pattern (CSSP) filters with the Tunable-Q Wavelet Transform (TQWT) to extract highly discriminative features from EEG signals while reducing the impact of individual-specific

variations. CSSP is used to optimize spatial filters that enhance the signal-to-noise ratio of MI-related EEG activity and TQWT provides a flexible time-frequency representation. To further enhance the robustness and generalizability of our method, we employ the Kruskal-Wallis non-parametric statistical test for rigorous selection of the top 10 most discriminative features. This feature selection step ensures that only the most relevant and informative features are used for classification, reducing the risk of overfitting to individual-specific characteristics. Our method is evaluated on the publicly available CHO-2017 database [14], a well-established benchmark in MI research. Using a K-Nearest Neighbors (KNN) classifier on these selected features, our proposed method achieved an exceptional average classification accuracy of 98.84%. This superior performance demonstrates the effectiveness of our hybrid signal processing and feature selection strategy in enhancing the accuracy, robustness, and generalizability of MI-BCI systems.

The structure of this paper is as follows: In the second part of the paper, the proposed method will be explained. In the third part, the results of using the proposed method on the CHO-2017 database will be presented, and the last part will be related to the conclusion.

II. PROPOSED METHOD

A. Dataset

A significant challenge in BCI research is understanding and mitigating inter-individual variability in MI task performance. Developing algorithms for MI recognition that are robust and independent of the specific user is a key objective in this field. For this study, we utilized the CHO-2017 database, meticulously collected and curated by Cho et al. [14]. This comprehensive dataset includes not only EEG recordings but also subjective responses from psychological

and physiological questionnaires, providing a rich context for our analysis.

The experiments in the CHO-2017 database were designed to capture neural activity related to the perception of right and left hand movements. The study involved 52 participants (19 women, 33 men) with a mean age of 24.8 years (standard deviation: 3.86). The majority of participants were right-handed, with participant #33 being non-dominant. Ethical approval for the experiments was obtained from the Guangzhou Institute of Science and Technology Review Board, and all participants provided written informed consent and received monetary compensation. EEG data were acquired using a Biosemi Active Two system equipped with 64 active Ag/AgCl electrodes, positioned according to the 10-10 international system. The data were sampled at 512 Hz. The BCI2000 3.0.2 system facilitated the collection of EEG data and the presentation of MI task instructions. To differentiate between actual hand movements and imagined ones, EMG data were recorded simultaneously with EEG using two electrodes placed on the flexor digitorum profundus and extensor digitorum muscles of each arm. The precise 3D coordinates of the EEG channels were meticulously measured using a Polhemus Fastrak 3D coordinate digitizer.

A distinctive feature of the CHO-2017 database is the inclusion of auxiliary data types beyond the primary MI tasks. These include recordings of blinking, upward/downward and left/right eyeball movements, head movements, jaw movements, and rest states. Each of these auxiliary movements was recorded twice for 5 seconds, while the rest position was recorded for 60 seconds. These ancillary recordings are invaluable for noise removal and artifact rejection processes.

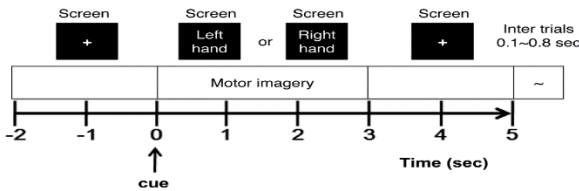


Fig. 2. MI test instructions [15].

In typical cognitive science research, biological data recordings often occur concurrently with task performance. Before commencing the MI tasks, participants were instructed to perform a specific preparatory movement: touching the index finger to the thumb within 3 seconds. For the MI trials, participants were guided to engage in kinesthetic imagination rather than visual imagination of the movements. During trials involving actual hand movements, participants sat with armrests and viewed a monitor. Each experimental trial commenced with a 2-second display of a black screen with a fixed crosshair, signaling the participant to prepare. Following this, a cue indicated whether to imagine moving the left or right hand for 3 seconds, as randomly presented on the screen (See Fig. 2). After the imagined or actual movement, a blank screen reappeared, followed by a randomized rest period ranging from 0.25 to 0.8 seconds. This sequence constituted one trial. A “run” consisted of 100 repetitions for a single class of movement imagery (e.g., only left hand MI).

Another noteworthy aspect of this database is the accompanying questionnaire, administered at different stages of the experiment. Participants responded to 15 questions before the experiment (questions 101-115), 10 questions after

each run (questions 210-219), and the remaining questions after the entire experiment concluded (questions 301-304) [25]. This structured collection of subjective feedback can provide valuable insights into participants’ experiences and potential influences on their performance..

B. Preprocessing

EEG signals, characterized by their low amplitude, are inherently susceptible to various forms of noise. Consequently, the preprocessing stage is critical for isolating the relevant neural information from extraneous artifacts. A primary objective is the removal of ocular artifacts, such as Electrooculogram (EOG) signals and blinks, which can significantly contaminate the raw EEG data. To address this, we employed Independent Component Analysis (ICA), a widely recognized and effective technique available within the EEGLAB toolbox for MATLAB. ICA decomposes the multi-channel raw EEG data into a set of statistically independent components (ICs). Each IC is then visually inspected and labeled based on its characteristic topography and time-course to identify those representing artifacts (e.g., blinks, eye movements). Components identified as artifactual are subsequently removed, and the remaining ICs are recombined to reconstruct a cleaner EEG signal.

Furthermore, the frequency characteristics of EEG signals relevant to Motor Imagery (MI) tasks are primarily concentrated within a specific range. Building upon findings from previous studies, which suggest that EEG activity outside the 8–30 Hz band contains minimal useful information for MI classification, we applied a 5th-order Butterworth bandpass filter. This filtering step effectively attenuates frequencies outside this informative range, thereby removing irrelevant spectral content and enhancing the signal-to-noise ratio for MI-related brain dynamics.

Finally, recognizing the inherent trade-off between temporal and spatial resolution in EEG, we implemented a spatial Laplacian filter. This technique aims to improve source localization by emphasizing localized brain activity and reducing the influence of distant, potentially irrelevant, neural sources, consistent with the methodology described in [16]. This final step helps to refine the spatial characteristics of the processed EEG signals prior to feature extraction.

C. Common Spatial-Spectral Pattern

CSP was first proposed for the purpose of classifying multichannel EEG signals during hand movement perception [17]. Its main idea is to use a linear transformation to map multichannel EEG data into a lower-dimensional space with a mapping matrix where each row represents weights for the channels. This transformation is based on the simultaneous diagonalization of the correlation matrix of two distinct classes [18].

Let X_1^i and X_2^i be the EEG data matrices of two arbitrary classes of movement perception in the i th trial with dimensions $N \times T$. N denotes the number of channels and T represents the number of sample points in time. Then the normalized correlation matrices corresponding to each class are obtained from equation (1).

$$R_1^i = \frac{X_1^i X_1^{iT}}{\text{trace}(X_1^i X_1^{iT})}, \quad R_2^i = \frac{X_2^i X_2^{iT}}{\text{trace}(X_2^i X_2^{iT})} \quad (1)$$

The normalized correlation matrix $\bar{R}$ is calculated by averaging over all the trials of each class.

$$1 \quad \frac{1}{N_1} \quad \frac{N_1}{i=1} \quad \frac{i}{1} \quad , \quad 2 \quad \frac{1}{N_2} \quad \frac{N_2}{i=1} \quad \frac{i}{2} \quad (2)$$

After calculating these two matrices, the composite correlation matrix can be obtained using equation (3).

$$R_c = \bar{R}_1 + \bar{R}_2 \quad (3)$$

Furthermore, the eigenvalue decomposition of the matrix R_c can be calculated from equation (4).

$$R_c = B_c \lambda B_c^T \quad (4)$$

The matrix B_c contains the normalized eigenvectors R_c and λ is the diagonal matrix of the eigenvalues of R_c . Both of these matrices are of dimension $N \times N$. Now, using equations (5), (6) and (7), the matrices S_1 and S_2 are obtained. These matrices are the whitened versions of R_1 and R_2 .

$$P = \lambda^{-1/2} B_c^T \quad (5)$$

Whitening is used to remove any (linear) dependence between the components of the correlation matrix.

$$S_1 = P \bar{R}_1 P^T \quad (6)$$

$$S_2 = P \bar{R}_2 P^T \quad (7)$$

S_1 and S_2 share the same common eigenvectors, which are obtained by using the eigen factor of relations (8) and (9).

$$S_1 = U \psi_1 U^T \quad (8)$$

$$S_2 = U \psi_2 U^T \quad (9)$$

It should also be noted that $\psi_1 + \psi_2 = I$, where I is the identity matrix. The weight matrix W can be calculated according to relation (10) using the common eigenvectors and the whitening matrix P .

$$W = U^T P \quad (10)$$

Since the sum of two corresponding eigenvalues is always equal to one, the eigenvector with the largest eigenvalue for S_1 will have the smallest eigenvalue for S_2 , and vice versa. This property makes the eigenvectors U very useful for classifying two distributions. Accordingly, the first and last m columns of the weight matrix W (each column representing a filter) provide the most discriminating features between the two classes. Now, suppose that $\hat{W}$ is a matrix consisting of only the first m columns and the last m columns of the matrix W . After the original EEG data are mapped to the new space using $\hat{W}$, the discrimination between the two classes of EEG data will be maximized [18]. Using the matrix $\hat{W}$ and considering the relation (11), the data related to each trial (X^i) can be mapped to the new space [18].

$$Z^i = \hat{W} X^i \quad (11)$$

In the above equation, X^i denotes the data matrix of the i -th trial and Z^i denotes the data matrix mapped to the new space for the i -th trial. The variable m , which is the number of CSP filters, is a meta-parameter in this algorithm that is set by trial and error and empirically based on the characteristics of each database. By executing the CSP algorithm, the data matrices X_1^i and X_2^i with dimensions $N \times T$ will be mapped to two matrices Z_1^i and Z_2^i with dimensions $2m \times T$. In this research, we have used an improved version of CSP called CSSP, which we will discuss in detail below.

Suppose we denote the j th row of the data matrix X_1^i by $x_j^i(n)$. In the CSSP algorithm, the signal $x_j^i(n)$ is filtered with a time delay τ_n by the pass-through FIR filters $h_k = [h_k(0), h_k(1), \dots, h_k(L-1)]$ $k = 1, \dots, N_f$ in different frequency bands:

$$\hat{x}_{j,k}^i(n) = \sum_{l=0}^{L-1} h_k(l) x_j^i(n-l-\tau_n) \quad (12)$$

Now suppose that the data matrices $\hat{X}_{1,k}^i$ and $\hat{X}_{2,k}^i$ are the result of applying the row-by-row filtering operations of equation (12). To obtain the mapping matrices $\hat{Z}_{1,k}^i$ and $\hat{Z}_{2,k}^i$, the steps in Equations 1 to 11 are performed for each of the data matrices $\hat{X}_{1,k}^i$ and $\hat{X}_{2,k}^i$. The choice of the number of filters, frequency bands, and time delay depends on the application and type of application. Another way to implement the CSSP algorithm without the need to design h_k filters is to add time-delayed versions of the data matrices $\hat{X}_{1,k}^i$ and $\hat{X}_{2,k}^i$ to themselves as follows:

$$\hat{X}_1^i = \begin{bmatrix} X_1^i \\ \delta^\tau X_1^i \end{bmatrix}, \hat{X}_2^i = \begin{bmatrix} X_2^i \\ \delta^\tau X_2^i \end{bmatrix} \quad (13)$$

where δ^τ is the delay operator and is defined as $\delta^\tau(X_{0,t}) = X_{0,t-\tau}$. Each time delay produces a phase shift in the frequency domain, the combination of these delayed versions of the signal in the final filter created by CSSP acts similarly to the same h_k band-pass filters [19].

In this study, the CSSP algorithm was first applied to the raw EEG signals to derive spatial filters W that maximize discriminability between the two motor imagery classes. These filters were calculated strictly from the training data in each fold of the subject-specific 5-fold cross-validation, ensuring no contamination from test data. The spatial filters were then applied to both training and test EEG signals, reducing the original multi-channel signals (e.g., 64 channels) to a smaller set of spatially filtered components (e.g., 12 components). Subsequently, the TQWT was performed on each spatially filtered component to decompose the signals into several frequency sub bands. From these sub bands, statistical features including variance, skewness, kurtosis, and energy were extracted. This sequential processing pipeline — spatial filtering followed by wavelet-based feature extraction — was performed independently on both training and test datasets to maintain rigorous validation integrity.

D. TQWT

The Quality Factor (Q) in wavelet transforms is a key parameter that governs how well wavelets can analyze both frequency and time characteristics. It directly influences a wavelet's ability to distinguish between specific frequencies and its precision in time localization. Essentially, Q is the ratio of a wavelet's center frequency to its bandwidth. A high Q signifies a narrow frequency band, allowing for more accurate separation of distinct frequencies, though it may reduce time resolution. Conversely, a low Q indicates a wider bandwidth, which can capture a broader range of signal features and offers better time localization but with less accuracy in frequency separation. The optimal Q is ideally chosen to match the signal's oscillatory nature. For instance, analyzing non-stationary oscillatory signals like speech or EEG typically requires a high Q , while processing signals with minimal oscillations benefits from a low Q . Unlike the continuous wavelet transform, most other wavelet transforms offer limited adjustability for Q . To overcome this, the TQWT was developed for discrete-time signals, featuring an easily adjustable Q factor and an oversampling rate [20]. The TQWT utilizes a multi-rate filter bank (as seen in Fig 3), generating low-pass ($v_0(n)$) and high-pass ($v_1(n)$) sub-band signals at

sampling rates of f_s and f_s , respectively, where f_s is the input signal's sampling frequency.

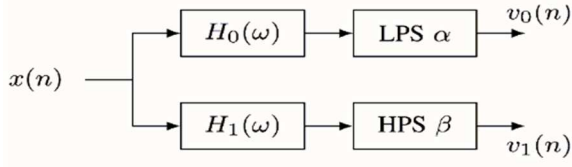


Fig. 3. TQWT analysis filter bank for one stage [20].

In discrete wavelet transform, the value of Q is fixed, but in TQWT, this parameter is adjustable and independent of level j , and depends only on the filter bank β . The two-channel filter bank shown in Fig. 3 is oversampled by $\alpha + \beta$. If we continue this filter bank from its low-pass output and repeat it, it is as if we have implemented a wavelet transform with an oversampling rate or redundancy of $r = \frac{\beta}{1-\alpha}$. The reason for this is that below the j th level, the sampling rate is $\beta \alpha^{j-1} f_s$. As a result, the sum of the sampling rates of all levels together will be $\frac{\beta}{1-\alpha} f_s$.

Usually, in practice, when using TQWT transform, we first pre-select our desired Q and r and, based on equation (14) and the redundancy relation $r = \frac{\beta}{1-\alpha}$, calculate the values of α, β as follows and implement the TQWT filter bank based on them:

$$Q := \frac{\omega_c}{BW} = \frac{2 - \beta}{\beta} \quad (14)$$

$$\alpha = 1 - \frac{\beta}{r}, \quad \beta = \frac{2}{Q + 1} \quad (15)$$

Finally, it should be noted that Q and r are not necessarily integers. The value of Q should be set so that $Q \geq 1$, setting Q to a value of 1 will result in a wavelet transform whose wavelet will be similar to the second derivative of a Gaussian function, higher values of Q will produce wavelets with more oscillations. The value of r should also be definitely greater than 1. Setting the value of r to 1 will result in the generation of a sinc-like wavelet in the time domain. To avoid this problem, it is sufficient to choose $r \geq 3$. In this case, the passband of the frequency response of the j th level will not be flat [20].

E. Feature Extraction

After extracting the frequency sub-bands from each CSSP output channel, the features in equations (16) to (24) are extracted from each of them and the feature vector is formed:

$$\text{mobility} = \sqrt{\frac{\text{Var}\left(\frac{dx(t)}{dt}\right)}{\text{Var}(x(t))}} \quad (16)$$

$$\text{Complexity} = \frac{\text{mobility}\left(\frac{dx(t)}{dt}\right)}{\text{mobility}(x(t))} \quad (17)$$

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (18)$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \quad (19)$$

$$\text{skewness} = \frac{1}{N} \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3} \quad (20)$$

$$\text{kurtosis} = \frac{1}{N} \sum_{i=1}^N \frac{(x_i - \mu)^4}{\sigma^4} \quad (21)$$

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (23)$$

$$STD = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}} \quad (24)$$

$$MAD = \text{Median}(|x_i - \text{Median}(x_i)|) \quad (24)$$

$$\text{Total_Power} = \frac{1}{N} \sum_{i=1}^N x_i^2 \quad (25)$$

III. RESULTS

In this study, 5-fold cross-validation was used to evaluate the performance of the classifiers in the proposed algorithm, and then their average was reported. To mitigate overfitting and prevent data leakage, a stringent subject-specific 5-fold cross-validation was employed. For each subject, all trials were randomly and stratified split into five folds, with 80 trials per class assigned for training and 20 trials for testing in each fold, maintaining class balance. Spatial filters and feature extraction parameters were exclusively derived from the training folds, and then separately applied to the testing folds. This protocol was repeated across five independent runs, and final performance metrics are reported as the average accuracy.

The results are reported using the evaluation criteria of accuracy, precision, sensitivity, specificity, and F1 criterion. It should also be noted that the results of each classification after 5 runs for each individual are reported as the average of these results. The computational framework of this paper was performed on a computer system with an 11th generation Intel Core i5-11400F processor and a base frequency of 2.60 GHz. This system has a DDR4 RAM with 16 GB of memory. This execution was performed on MATLAB software version 2023b.

In this study, the values of the parameters required to implement the CSSP algorithm were selected empirically. Many experiments were conducted to obtain the appropriate number of spatial filters (m) for the CSSP algorithm. The criterion for selecting m was to obtain high accuracy in motion image recognition for different classifiers. The performance of the proposed method was evaluated for values of $m=3, \dots, 20$. The experimental results showed that for the aforementioned database, $m=6$ provides acceptable results for all 52 individuals in the experimental database. Thus, we selected 6 first and 6 last filters, for a total of 12 filters, to be applied to our 64-channel data. It should be noted that the performance of the proposed method was also investigated for different values of the delay parameter and its appropriate performance was observed for $\tau=4$.

Table 1. Frequency bands of TQWT decomposition up to 15 steps for $Q=2.4, r=3$

Level of decomposition	Frequency band (Hz)	Overlap with important EEG rhythms	Level of decomposition	Frequency band (Hz)	Overlap with important EEG rhythms
D9	9.19-22.33	-alpha beta	D1	52.70-128	gamma
D10	7.39-17.95	-alpha beta	D2	42.37-102.90	gamma
D11	5.94-14.43	-theta -alpha beta	D3	34.06-82.72	gamma
D12	4.7-11.60	-theta	D4	27.38-66.50	-beta

		alpha			gamma
D13	3.84-9.32	-theta alpha	D5	22.01-53.46	-beta gamma
D14	3.08-7.49	-delta theta	D6	17.69-42.98	-beta gamma
D15	2.48-6.02	-delta theta	D7	14.22-34.55	beta
A15	1.99-4.84	delta	D8	11.43-27.77	beta

As mentioned earlier, the value of r or the additional sampling rate ensures that the separated energy subbands are well preserved and more accurate reconstruction is performed. After applying CSSP, TQWT transformation is applied to each of the output channels of this algorithm. The values of the parameters Q and r were also set to 2.4 and 3, respectively. Choosing Q in the range of 2 to 3 increases the accuracy in separating the EEG bands. Thus, the values of α and β were equal to 0.804 and 0.588, respectively. The number of decomposition steps (j) was chosen to be 15 based on the choice of the values of Q and r and also the sampling frequency of 512 Hz of the database used. The reason for choosing 15 decomposition steps is shown in Table 1. This table shows the frequency ranges that the approximation coefficients and details of the TQWT transformation provide for a sampling frequency of 512 Hz. As can be seen, by choosing $j=15$, we can ensure that the frequency bands obtained from the decomposition will have a good overlap with the important frequency bands of the brain rhythms. According to Table 1, we find that the sub-bands extracted from steps 1 to 6 of the TQWT decomposition include information outside the frequency range between 8 and 30 Hz or repetitive information due to overlapping frequency ranges. Therefore, no feature is extracted from these sub-bands (D1-D6).

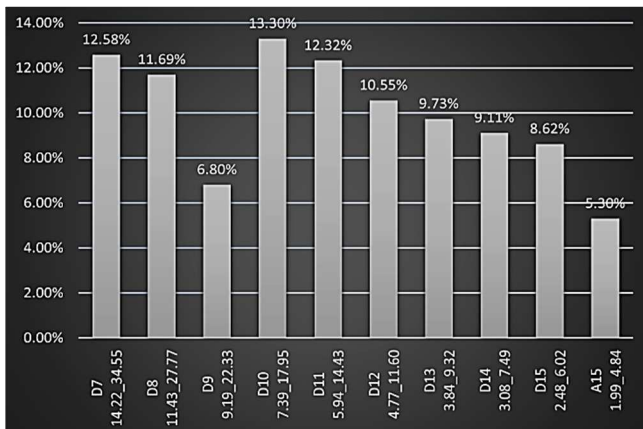


Fig. 4. Ranking sub-bands based on the best features extracted from each sub-band in the proposed approach for different people.

After extracting the features, the features were ranked based on the p -value calculation using the Kruskal-Wallis test. Based on the ranked features, it can be investigated what share of these top features each sub-band of the TQWT analysis has allocated to itself, assuming that the number of selected top features is kept constant (for example, top 100 features). As can be seen in Fig 4, it can be concluded that about 65% of the top features belong to the frequency range related to the motion perception reported in the research. According to the

results, the sub-bands D10, D7, D8 have obtained the best ranks in the frequency range in question, respectively, and the sub-bands D10, D7 play a fundamental role for most people.

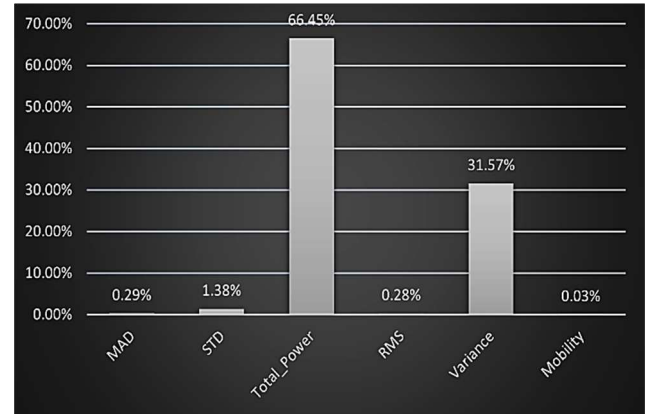


Fig. 5. Ranking top features in the proposed approach for different people.

Also, using this test, it is possible to identify the type of influential features among the top 100 features. Figure 5 shows six categories of features that have a significant contribution among the top features. This figure shows that the feature “mean sum of squares of signal wavelet transform coefficients”, which is a measure of the total energy of the signal in the time-frequency domain, has the best performance in separating the two classes with a contribution of 66.45%. After that, the variance feature with a contribution of 31.57% shows the second best performance in separating the classes. These results indicate that in the proposed method, these two features (especially the first feature) are efficient for all

Table 2. Classification results based on the proposed approach using the top 10 features.

F1 Score	Specificity	Sensitivity	Precision	Accuracy	Classifiers
98.17	96.79	98.94	97.82	97.86	RBF-SVM
98.28	98.73	97.81	98.87	98.27	Polynomial-SVM
98.89	98.70	98.99	98.81	98.84	KNN
96.5	92.55	97.87	95.18	95.21	DT
95.67	98.39	93.66	98.64	96.02	LDA

individuals and classification can be performed solely based on these two features. Given these achievements, we were confident that classification with a much smaller number of features is also possible. Our prediction was confirmed and we were able to achieve satisfactory results using only the top 10 features. Table 2 shows the results of implementing the proposed method in the classification of right and left hand movement perception. These results are based on the average classification performance of 52 individuals in the training database. As can be seen in Table 2, the two classifiers KNN and Polynomial-SVM, with an average accuracy of 98.84 and 98.27 percent, have provided the best performance among the different classifiers for the separation of right and left hand movement perception tasks.

Table 3. Comparison of the proposed algorithm with other studies conducted on the common database.

Accuracy (%)	Description	Year	Methods
65.02	Deep Learning, Using Adversarial CO-learning Networks	2024	Liu et al.[21]
66.69	Deep learning, using the Phase-SPDNet method	2024	Carrara [10] et al.
75.18	Supervised Contrastive Learning-Based Domain Generalization Network (SCLDGN)	2024	Xi et al.[11]
76.55	Transfer learning includes joint distribution adaptation (JDA) and Riemannian direction-preserving transfer learning (RLPTL)	2024	Zhou et [12] al.
77.03	D Generative Competitive 3 Convolutional Neural Network (3D-CNN-GAN)	2024	Fan et al. [13]
69.85	Multi-Source Deep Domain Group Adaptation Framework (MSDDAEF)	2024	Miao et [22] al.
98.84	Combination of CSSP and TQWT	2024	Proposed Method

To prove the effectiveness of the proposed approach in classifying motor imagery tasks, the performance of this method was compared with other modern methods based on deep learning and more complex algorithms that were implemented on the same database, and the results are presented in Table 3. The results in Table 3 show that our proposed method has achieved a significant improvement in classification accuracy compared to previous studies conducted on the aforementioned database. The best previously reported accuracy was close to 78%, while the proposed method has achieved an accuracy exceeding 98%. This high accuracy was achieved even despite the relatively poor performance of the proposed method in classifying data related to person number 33; a case for which other methods also failed to provide acceptable results. Further investigation showed that this person does not have a dominant hand and is able to use both hands with similar quality and power. According to the analysis of the superior features in the previous chapters, the reason for the poor classification performance for this individual may lie in the fact that the superior feature is the total energy of the frequency components in a specific time interval; while this individual, due to the lack of dominant hand, activates both hemispheres of the brain (responsible for controlling the right and left hands) with close power. Therefore, this feature does not perform well for separating classes in individuals with such unique characteristics. Finally, it can be concluded that solving the challenge of interpersonal variability is the key to developing brain-computer interface (BCI) systems with higher accuracy and efficiency; a challenge that has been carefully considered in this research and considerable effort has been made to solve it.

IV. CONCLUSION

In this study, by combining CSSP filters and TQWT transform, an effective approach for classifying left and right hand motor imagery tasks was presented. Using only the top ten extracted features, while reducing the complexity of the algorithm, achieved a high accuracy of 98.84% with the KNN classifier. While simple statistical features and KNN classifiers are conventional, our hybrid approach integrating CSSP and TQWT offers a unique advantage. CSSP adaptively optimizes spatial filters to each subject's EEG patterns, effectively addressing inter-subject variability prevalent in

motor imagery tasks. TQWT, with its tunable frequency resolution, facilitates adaptive time-frequency decomposition, enhancing feature discrimination. Comparative experiments included in the revised manuscript demonstrate that direct feature extraction from raw or frequency-filtered signals yields inferior classification performance, thereby justifying the necessity of combined spatial-spectral processing for improved accuracy. The notable improvement over traditional CSP and recent deep learning approaches stems from this carefully tuned integration rather than methodological novelty alone. The results also showed that the total signal energy and variance features play a key role in distinguishing between classes. In addition, the studies conducted on inter-individual variability confirmed the importance of addressing this challenge for developing brain-computer interface systems with greater accuracy and stability. The proposed method can be used as a strong foundation for developing clinical and practical applications in the field of BCI.

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